

Aditya Kulkarni

(281) 876-7066 | adityakulkarnius@gmail.com | linkedin.com/in/aditya-kulkarni-355b81217 | github.com/Aditya-k24 | kulkarniaditya.com |
Raleigh, NC

PROFESSIONAL SUMMARY

MS CS candidate at NC State (May 2026) with production experience shipping AI-powered backend systems, LLM pipelines across 3 providers, and full observability stacks. Delivered 10+ features end-to-end at an AI startup across 7 deployments. Published IEEE researcher and top-10 national hackathon finalist.

WORK EXPERIENCE

Isomer AI May 2025 – Present

Product Engineer Intern

- Ship 10+ production backend features using TypeScript, Node.js, Express, PostgreSQL, Redis, Docker, and Terraform across 7 deployments, including role-based access control, real-time event pipelines, and Temporal.io workflow orchestration.
- Integrate multi-model AI agent tooling across 3 providers and 8+ models (Anthropic Claude, OpenAI GPT-4o, Google Gemini) via MCP, building provider-agnostic interfaces with structured outputs and document processing pipelines (PDF generation, email parsing, S3).

Mphasis (HP Company) Jun 2024 – Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML, improving claim processing accuracy by 25%.
- Automated cloud deployments on AWS and GCP, integrated Grafana monitoring dashboards across production services, and supported model red-teaming to improve system reliability.

Capgemini Jun 2023 – Aug 2023

AI/ML Intern

- Designed distributed data pipelines using Kafka, Spark, and Hadoop for real-time ingestion of support ticket data, reducing processing time by 15% and latency by 20%.

PROJECTS

LLM Eval Harness 2025

Built a vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini across 2,000 examples and 3 task types. Tracked hallucination rate, p50/p95 latency, and cost per call with bootstrap significance testing across 3 repeated runs per model.
Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

PulseRank – Real-Time Leaderboard & Activity Feed 2026

Engineered a real-time platform with full observability: OpenTelemetry traces and Prometheus metrics, MySQL outbox pattern, MongoDB TTL projections, Redis sorted sets for sub-millisecond reads, and Server-Sent Events for live updates. Load-tested with k6 and shipped via GitHub Actions CI/CD.
TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, OpenTelemetry, Prometheus, Docker, GitHub Actions

VectorLift – Semantic Search & Ranking Engine 2026

Built a semantic search system on MS MARCO evaluating 6 pipeline configurations: BM25 (Elasticsearch), dense bi-encoder retrieval (sentence-transformers), and cross-encoder re-ranker. Scored with NDCG@10, MRR@10, and MAP with bootstrap significance testing; served via FastAPI with per-stage latency tracking.
Python, PyTorch, Sentence-Transformers, Elasticsearch, Qdrant, Kafka, FastAPI, Docker

EDUCATION

MS Computer Science – North Carolina State University Aug 2024 – May 2026

AI, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering

B.Tech Computer Science – Mumbai University Aug 2020 – May 2024

CGPA 9.32/10 | Machine Learning, DS & Algorithms, Big Data Infra, Data Warehousing, Blockchain

SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, C, Bash

AI/ML: Anthropic Claude API, OpenAI API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, MCP, LLM Evaluation

Backend & Infrastructure: Node.js, Express, Spring Boot, FastAPI, Temporal.io, Kafka, Terraform, Docker, Kubernetes, AWS, GCP

Databases & Observability: PostgreSQL, Redis, MongoDB, Elasticsearch, OpenTelemetry, Prometheus, Grafana, Sentry

Awards: HackNC24 Top 10 finalist (100+ teams), 6+ national hackathon wins, IEEE Published (Springer 2023), Top 1% Sanjeevani health app (1100+ teams)